Genetic-Algorithm Optimization of Spreading Codes for Low Cross-Correlation

A concise literature and method summary — prepared for study/reference.

Why low cross-correlation matters

In DS-CDMA, GNSS, and MIMO-radar systems, each user or transmitter uses a spreading sequence. If sequences have high mutual correlation, signals interfere with one another (multiple-access interference), degrading acquisition, synchronization, and bit-error performance. A good code family therefore has a sharp autocorrelation peak at zero lag and low cross-correlation—and low out-of-phase autocorrelation—at all other lags.

How a genetic algorithm is applied

A GA treats each candidate code family as a chromosome: typically a binary matrix whose rows are the spreading sequences. The algorithm evolves a population through selection, crossover, mutation, and elitism. Its fitness function penalizes large off-peak autocorrelations and cross-correlations, often using mean-square correlation, integrated sidelobe level (ISL), or peak sidelobe level (PSL).

A common objective is to minimize the maximum of mean-square aperiodic autocorrelation and cross-correlation, or to minimize integrated sidelobe energy. Multi-objective variants such as NSGA-II preserve a Pareto front between low cross-correlation, good autocorrelation, and sequence-family size.

Typical GA formulation

1. Initialize N random binary sequences of length L.
2. Encode the set as a chromosome.
3. Compute periodic or aperiodic correlation for every sequence pair and lag.
4. Assign fitness, e.g. minimize average/maximum mean-square cross-correlation while constraining autocorrelation.
5. Select high-fitness parents; apply crossover and bit-flip mutation.
6. Retain elite solutions and iterate until the correlation metric converges.

Key papers and downloadable sources

Mina & Gao (2022), NAVIGATION	Natural Evolution Strategy vs. GA for GPS spreading codes	Minimizes max of mean-square auto/cross-correlation; outperforms Gold/Weil baselines for lengths up to 1023/1031 and 31-code families	https://navi.ion.org/content/69/1/navi.506
Mina & Gao (2019), multi-objective GA	High-performing random spreading-code sequences	Uses multi-objective GA to jointly reduce auto- and cross-correlation metrics	https://arxiv.org/html/2404.12629v1
Pradhan et al. (2019), IJITEE	NSGA-II for minimum-correlation spreading codes	Reduces out-of-phase mean-square aperiodic auto- and cross-correlation in random binary code sets	https://www.mendeley.com/catalogue/0698614a-fd8a-339e-9069-0766d0cfd62/
Boukerma et al. (2021), IJCS	Optimized long binary spreading sequences	GA and PSO optimize concatenated/interlaced Walsh-Hadamard and Gold subsequences; GA-optimized sequences improve ACF versus Gold, Weil, random codes	https://onlinelibrary.wiley.com/doi/abs/10.1002/dac.4719
River Publishers chapter	Spreading-code optimization using GA	Optimizes logistic-map-generated spreading codes; evaluates randomness and compares with DE-based optimization	https://www.riverpublishers.com/pdf/ebook/chapter/RP_P9788770040082C2.pdf

WSEAS (2014)	GA-based spreading-code design	Explicitly treats code-count versus peak cross-correlation as an optimization trade-off	https://www.wseas.com/journals/communications/2014/a025704-314.pdf

Practical takeaway

For a coursework or simulation project, start with random binary or Gold codes, then use a GA to minimize mean-square cross-correlation at all nonzero lags. Compare against unoptimized Gold/Walsh codes using peak cross-correlation, average mean-square cross-correlation, autocorrelation sidelobe level, and BER in an AWGN or multipath CDMA simulation. Published work shows GA and related evolutionary methods can improve correlation properties, although modern natural evolution strategies and convex/mixed-integer methods can be competitive or better for longer codes.

Note: The linked sources point to publisher, journal, preprint, or repository pages; availability of full-text PDFs depends on the source's access policy.